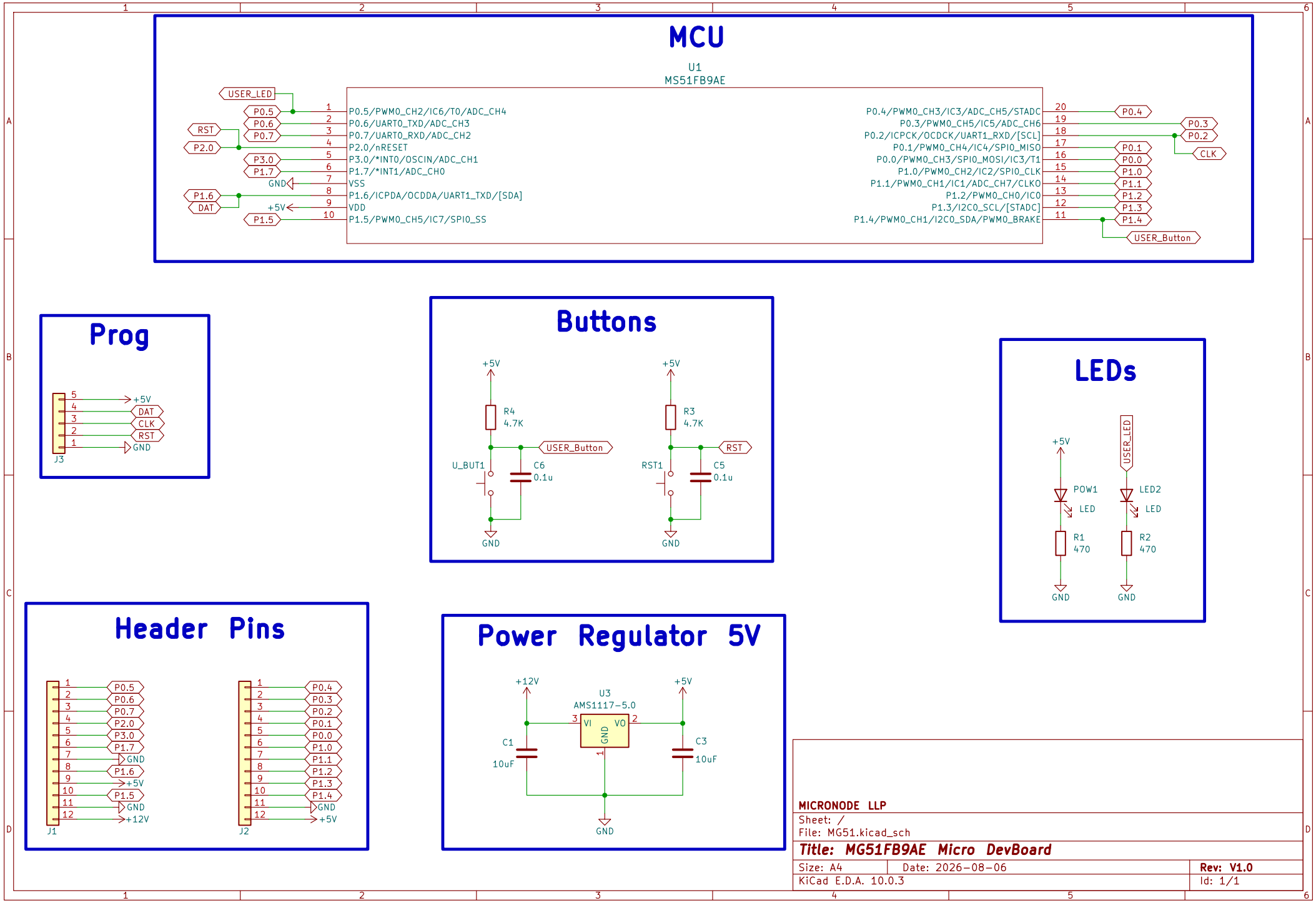

MCU
U1
MS51FB9AE
USER_LED
P0.5
P0.6
P0.7
RST
P2.0
P3.0
P1.7
GND
P1.6
DAT
+5V
P1.5
1 P0.5/PWM0_CH2/IC6/T0/ADC_CH4
2 P0.6/UART0_TXD/ADC_CH3
3 P0.7/UART0_RXD/ADC_CH2
4 P2.0/nRESET
5 P3.0/*INT0/OSCIN/ADC_CH1
6 P1.7/*INT1/ADC_CH0
7 VSS
8 P1.6/ICPDA/OCDDA/UART1_TXD/[SDA]
9 VDD
10 P1.5/PWM0_CH5/IC7/SPI0_SS
20 P0.4/PWM0_CH3/IC3/ADC_CH5/STADC
19 P0.3/PWM0_CH5/IC5/ADC_CH6
18 P0.2/ICPCK/OCDCK/UART1_RXD/[SCL]
17 P0.1/PWM0_CH4/IC4/SPI0_MISO
16 P0.0/PWM0_CH3/SPI0_MOSI/IC3/T1
15 P1.0/PWM0_CH2/IC2/SPI0_CLK
14 P1.1/PWM0_CH1/IC1/ADC_CH7/CLKO
13 P1.2/PWM0_CH0/IC0
12 P1.3/I2C0_SCL/[STADC]
11 P1.4/PWM0_CH1/I2C0_SDA/PWM0_BRAKE
P0.4
P0.3
P0.2
CLK
P0.1
P0.0
P1.0
P1.1
P1.2
P1.3
P1.4
USER_Button
Prog
5 +5V
4 DAT
3 CLK
2 RST
1 GND
J3
Buttons
+5V
R4
4.7K
USER_Button
U_BUT1
C6
0.1u
GND
+5V
R3
4.7K
RST
RST1
C5
0.1u
GND
LEDs
+5V
POW1
LED
R1
470
GND
USER_LED
LED2
LED
R2
470
GND
Header Pins
1 P0.5
2 P0.6
3 P0.7
4 P2.0
5 P3.0
6 P1.7
7 GND
8 P1.6
9 +5V
10 P1.5
11 GND
12 +12V
J1
1 P0.4
2 P0.3
3 P0.2
4 P0.1
5 P0.0
6 P1.0
7 P1.1
8 P1.2
9 P1.3
10 P1.4
11 GND
12 +5V
J2
Power Regulator 5V
+12V
U3
AMS1117-5.0
VI
VO
GND
3
2
1
C1
10uF
+5V
C3
10uF
GND
MICRONODE LLP
Sheet: /
File: MG51.kicad_sch
Title: MG51FB9AE Micro DevBoard
Size: A4
Date: 2026-08-06
Rev: V1.0
KiCad E.D.A. 10.0.3
Id: 1/1